C++ **manipulators** are special tools used with `cout` to control or format the output.

In this lesson, we will learn two commonly used manipulators:

1. `endl` – moves the cursor to the next line.
2. `setw` – sets the width of the output field.

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1. Manipulator: `endl`

What is `endl`?

`endl` is used with `cout` to:

- Move the cursor to the **next line**.
- Complete the current output line.
- Make the output easier to read.

Syntax

```
cout << "Text" << endl;
```

Example 1

Question: Write a C++ program that prints two lines of text using the manipulator `endl`.

Program

```
#include <iostream>
using namespace std;

int main() {

    cout << "Hello, C++ beginners!" << endl;
    cout << "This line is printed using endl.";

    return 0;
}
```

Output

```
Hello, C++ beginners!
This line is printed using endl.
```

Understand the Code

```
cout << "Hello, C++ beginners!" << endl;
```

- `cout` displays the text.
- `"Hello, C++ beginners!"` is the text to display.
- `endl` moves the cursor to the next line.

Therefore, the second `cout` statement starts printing from a **new line**.

Remember

`endl` is mainly used to **move output to the next line**.

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2. Manipulator: `setw`

What is `setw`?

`setw` means **set width**.

It is used to set the minimum width of the output field. It is especially useful when displaying data in **columns or tables**.

Important

`setw` is available in the `<iomanip>` library.

Therefore, we need to include:

```
#include <iomanip>
```

Syntax

```
setw(number)
```

For example:

```
cout << setw(10) << 25;
```

This gives the output field a width of **10 characters**.

Example 2: Using `setw` to Create a Table

Question: Write a C++ program that prints a simple table of numbers using `setw` to align the output in columns.

Program

```
#include <iostream>
#include <iomanip>

using namespace std;

int main() {

    cout << setw(10) << "Number"
         << setw(10) << "Square" << endl;

    cout << setw(10) << 2
         << setw(10) << 4 << endl;

    cout << setw(10) << 5
         << setw(10) << 25 << endl;

    cout << setw(10) << 10
         << setw(10) << 100 << endl;

    return 0;
}
```

Output

```
    Number    Square
         2         4
         5        25
        10       100
```

Understand the Code

Consider this statement:

```
cout << setw(10) << 2
     << setw(10) << 4 << endl;
```

- `setw(10)` gives the first value a field width of **10 characters**.
- `2` is displayed within that field.
- The second `setw(10)` gives the next value another field width of **10 characters**.
- `4` is displayed within the second field.
- `endl` moves the cursor to the next line.

This helps us display values in **aligned columns**.

Important Point About `setw`

`setw()` applies only to the **next output item**.

For example:

```
cout << setw(10) << 25 << 50;
```

Here, `setw(10)` applies to `25` only. It does not automatically apply to `50`.

If we want both values to have a width of 10, we write:

```
cout << setw(10) << 25
     << setw(10) << 50;
```

Quick Comparison

Manipulator	Purpose	Header File
`endl`	Moves output to the next line	`<iostream>`
`setw()`	Sets the width of the next output field	`<iomanip>`

Remember

- `endl` → **New line**
- `setw()` → **Set output width**
- `setw()` requires `<iomanip>`
- `setw()` affects only the **next output item**

Coding Exercises

Task 1 – `endl`

Write a C++ program that displays your:

- Name
- Department
- Semester

Use `endl` to print each item on a separate line.

Task 2 – `setw`

Write a C++ program to display the following table using `setw(10)`:

```
Number    Cube
2         8
3         27
4         64
5         125
```

Task 3 – Both Manipulators

Write a C++ program that uses both `endl` and `setw()` to display a simple student information table containing:

- Roll No.
- Name
- Marks